Devotopia

Verified Skill Roadmaps

A guided walkthrough of the product and the code behind it.
Every screenshot in this document was captured from the application running locally, and every figure is traced to a source file.

Graduation Project — Open Source Track — Alexandria

1. What Devotopia is

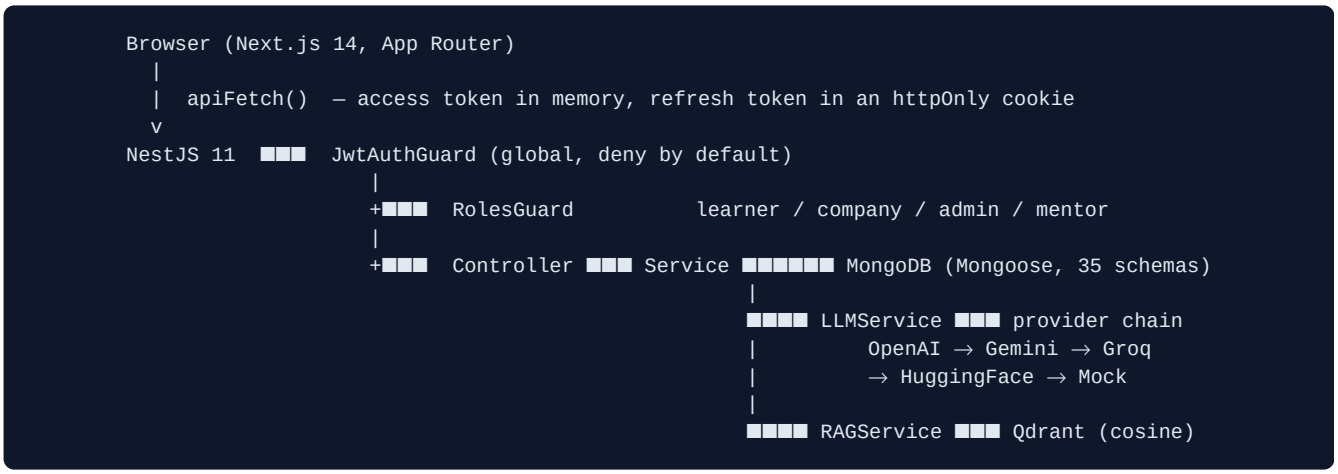
Devotopia turns a target job role into an ordered learning roadmap, then makes the learner prove each step. Every module is gated by an adaptive exam: pass it and the modules that depend on it unlock and a verified skill is written to the learner's passport; fail it and a shorter remedial module is generated from the exact questions that were missed. Companies read the same data in reverse, browsing candidates ranked by exam performance rather than by CV keywords.

The difference from a course catalogue is where the proof comes from. A completion certificate says a learner attended. A Devotopia verified skill says they answered questions at a measured difficulty and passed a weighted threshold.

The project in numbers

Repository layout	Turborepo monorepo, npm workspaces — apps/api, apps/web, packages/shared
Backend	NestJS 11 · 29 feature modules · 118 mapped HTTP routes · ~14,400 lines
Frontend	Next.js 14 App Router · 29 page routes · ~22,800 lines
Database	MongoDB via Mongoose · 35 schemas
Vector store	Qdrant, cosine similarity, collections verified on boot
AI layer	5 providers behind one fallback chain, ending in a deterministic mock
Languages	English and Arabic, with RTL layout switching and light/dark themes

How the pieces fit together

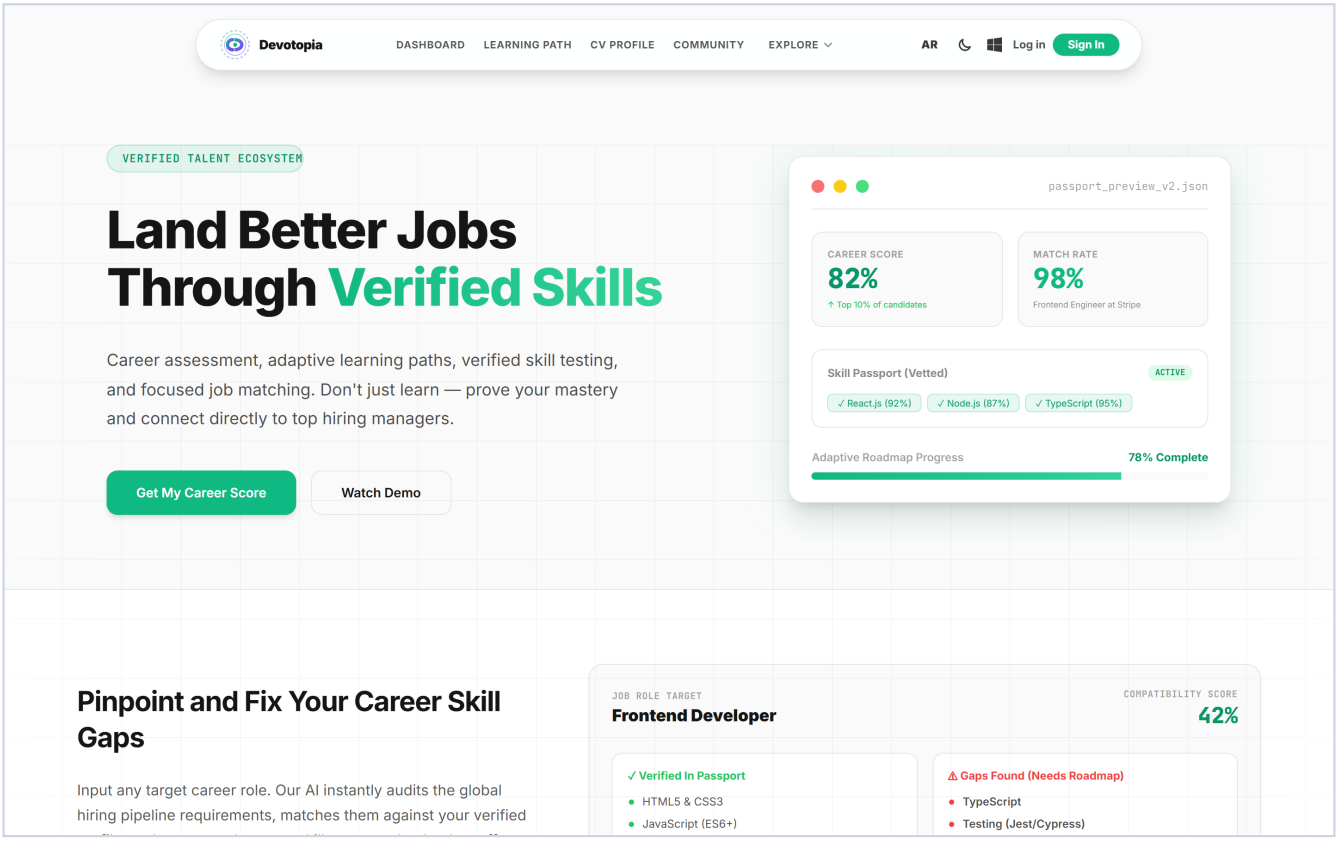


The request path. The guard runs before every controller unless a route opts out.

2. First contact — the public site

LANDING PAGE

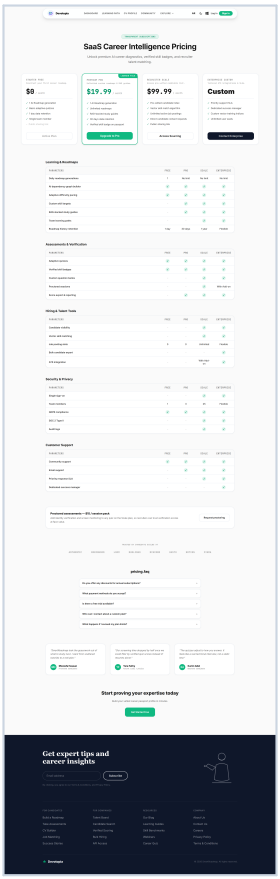
The entry point states the promise and routes visitors to registration. The navigation bar carries the language toggle (AR/EN), the theme switch, and the install prompt for the progressive web app — all three are present on every screen that follows.



The landing page at 1440×900. Language, theme and install controls sit in the navbar.

PRICING

Three tiers. The important detail is not on this page but behind it: the client never sends a price. It sends a plan name, and the server resolves the amount from a table before it creates the PayPal order, so a tampered request cannot buy a plan cheaply.



Pricing page. Plan amounts are resolved server-side in payment.dto.ts.

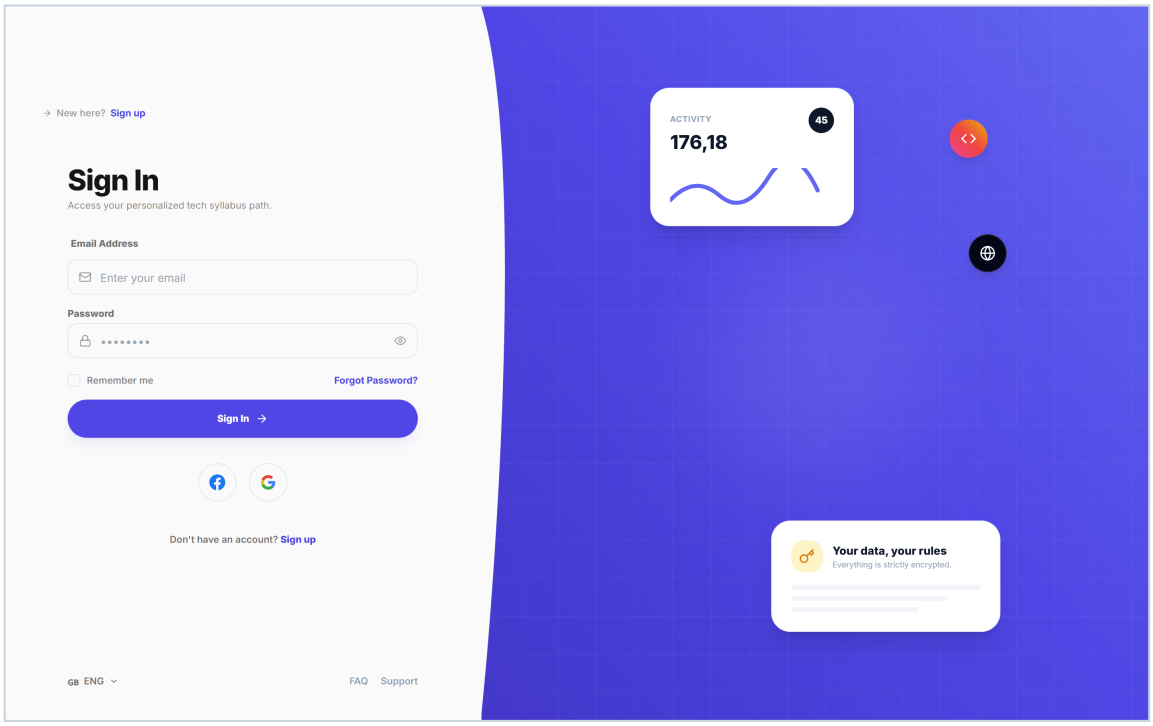
```
// apps/api/src/modules/payment/dto/payment.dto.ts
export const PLAN_PRICES: Record<'pro_learner' | 'company_tier', number> = {
  pro_learner: 19.99,
  company_tier: 99.99,
};

export class CreateOrderDto {
  // The price is resolved SERVER-side from this plan — the client never sends an amount.
  @IsIn(['pro_learner', 'company_tier'])
  plan!: 'pro_learner' | 'company_tier';
}
```

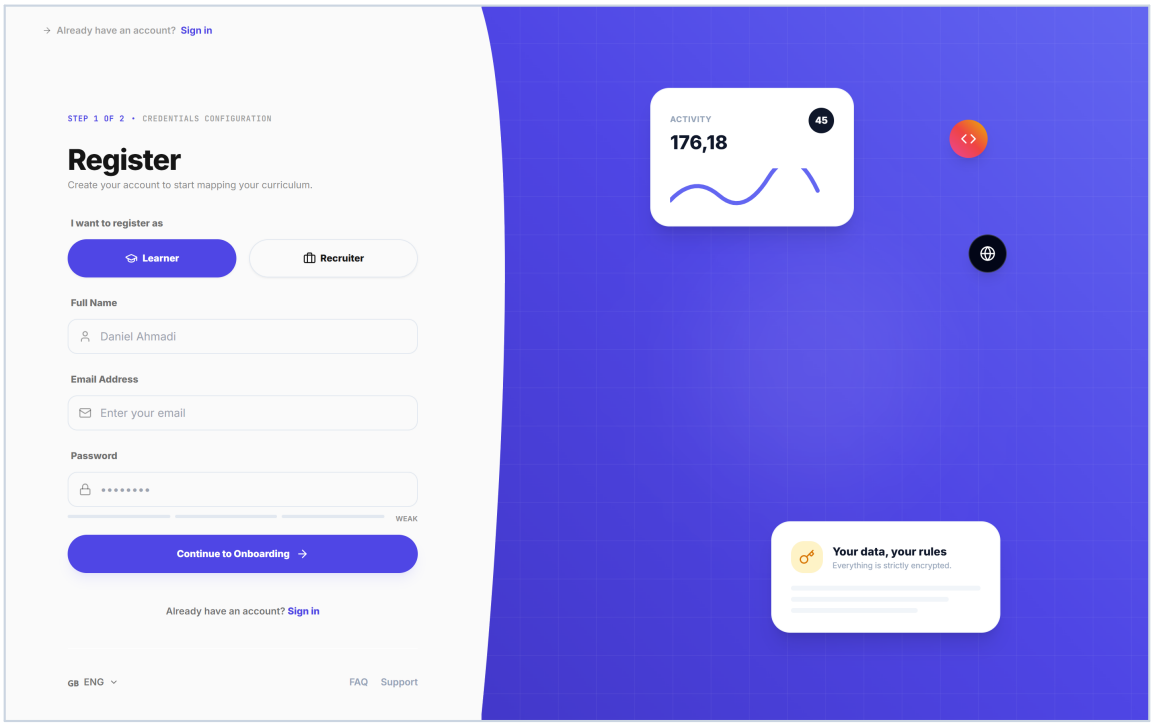
The comment in the source states the rule; the DTO enforces it by accepting only a plan name.

AUTHENTICATION

Registration and login share a token model built to survive an XSS incident. The refresh token is an httpOnly cookie scoped to /auth, so page JavaScript can never read it. The access token lives in a module variable and dies with the tab; on reload the app silently mints a new one from the cookie.



Login. The same screen handles Google identity sign-in.



Registration. Role is selectable, but 'admin' is rejected by the DTO.

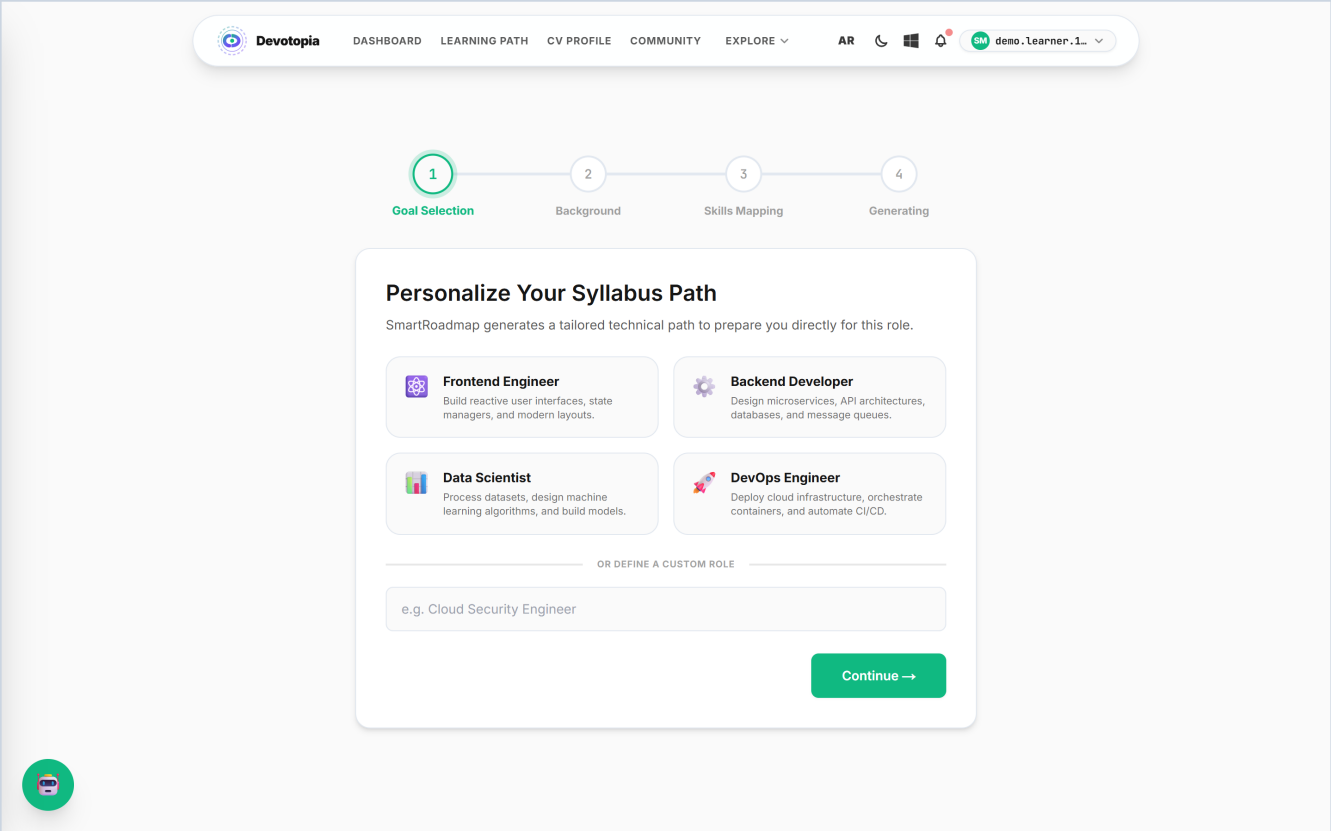
```
// apps/api/src/modules/auth/dto/auth.dto.ts
// NOTE: 'admin' is intentionally NOT assignable through public registration.
@IsOptional()
@IsIn(['learner', 'company', 'mentor'])
role?: 'learner' | 'company' | 'mentor';
```

Privilege escalation is closed at the validation layer, not in business logic.

3. The learner journey

STEP 1 — ONBOARDING

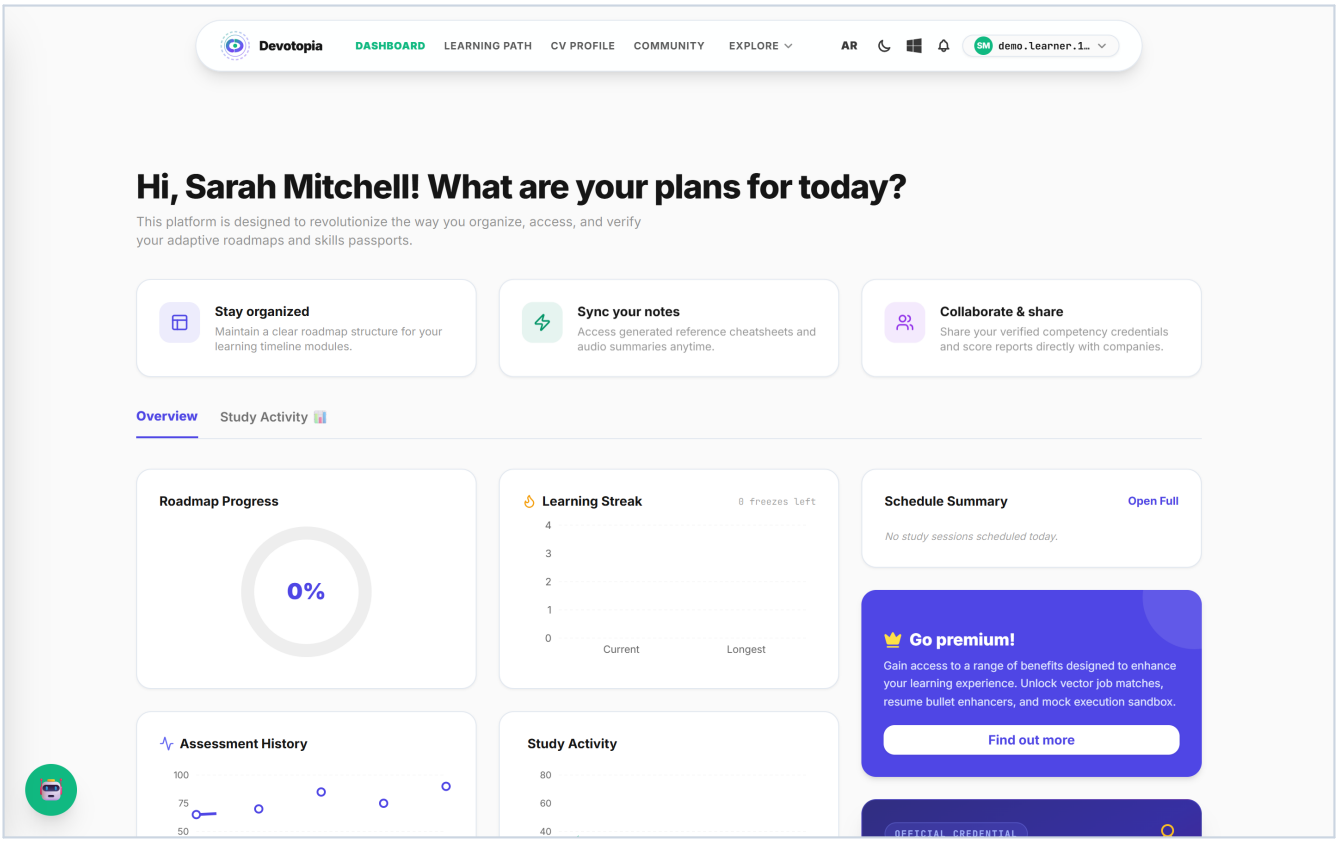
A short wizard collects the four inputs the curriculum generator needs: target role, education, years of experience, and current skills. Nothing else is asked, because nothing else changes the output.



Onboarding. These four answers become the prompt for roadmap generation.

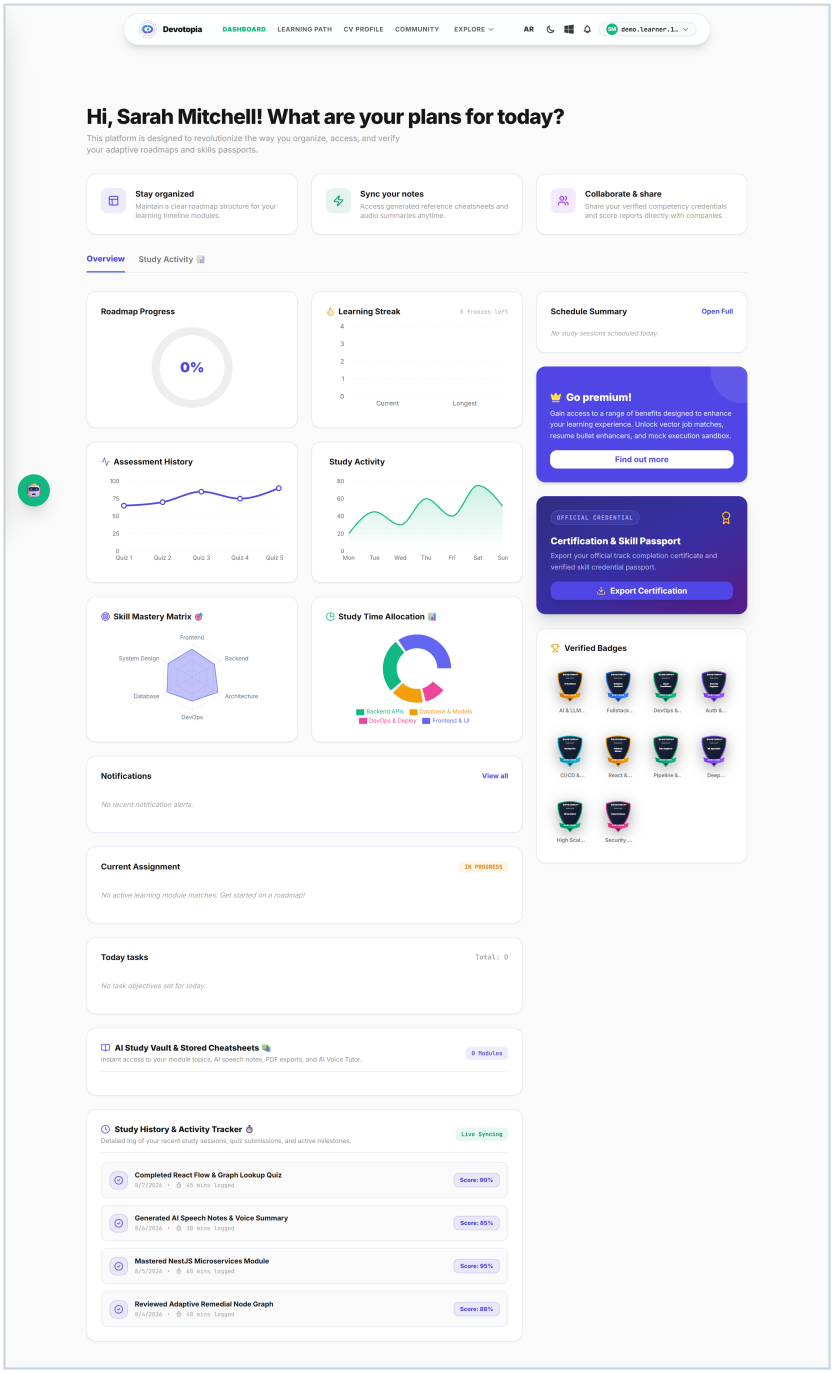
STEP 2 — THE DASHBOARD

After generation the learner lands here. Roadmap progress, the learning streak, assessment history and study activity are all live queries, not decoration — the streak and the achievement events are written by the same code path that finishes an exam.



Learner dashboard. Charts are Recharts, fed from the progress and streak modules.

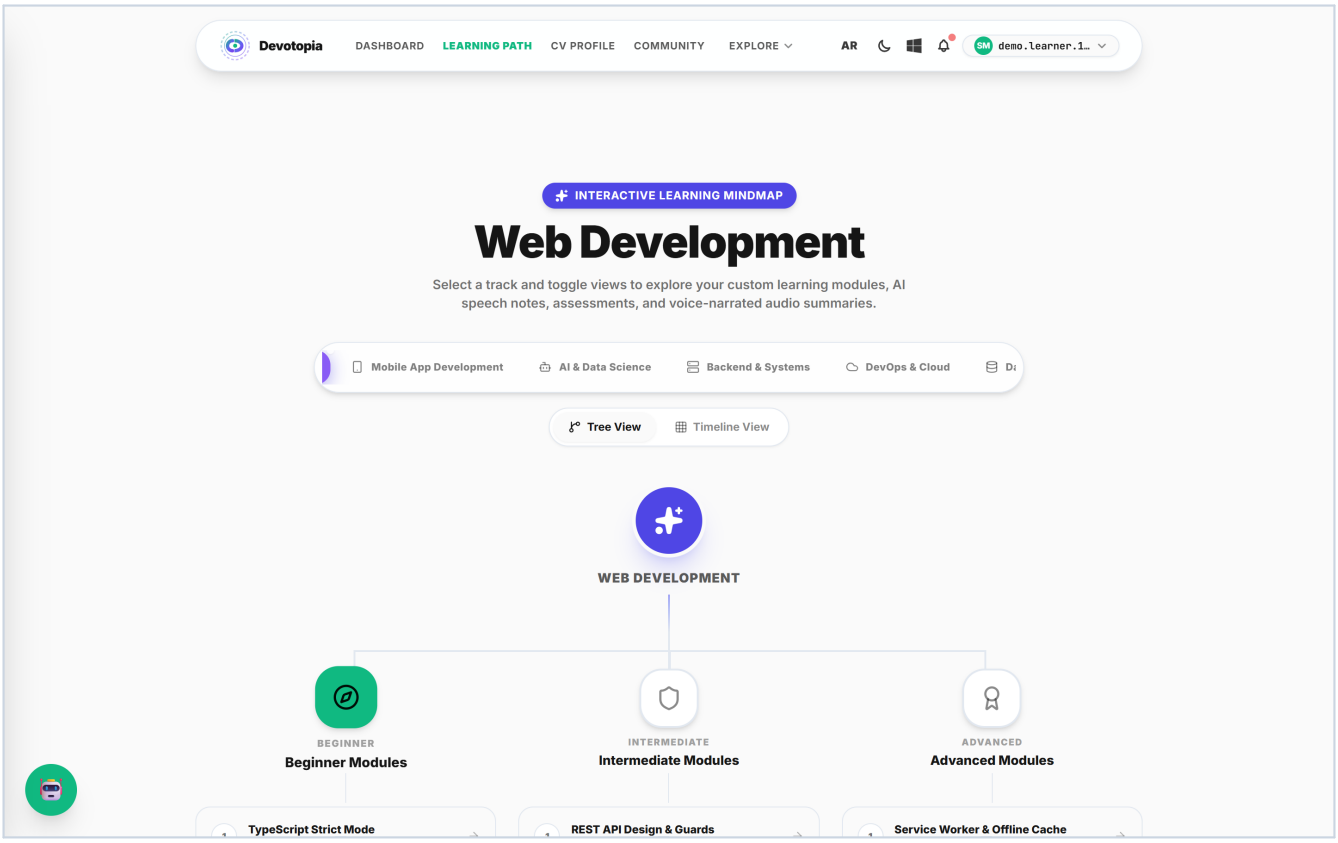
Scrolled out, the dashboard also carries the credential card and the recommended next actions:



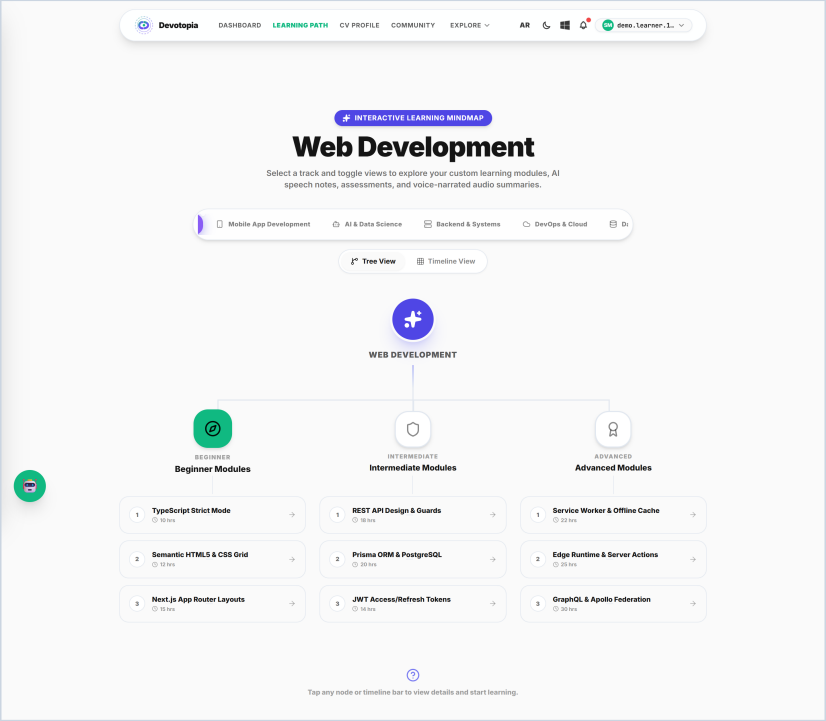
The full dashboard page.

STEP 3 — THE ROADMAP

The curriculum is rendered as a tree. Each node carries a difficulty tier, an hour estimate and a status — locked, in progress, completed or failed. The x/y coordinates for the canvas are produced by the generator itself, so the shape of the tree is part of the generated data rather than a layout guess made in the browser.



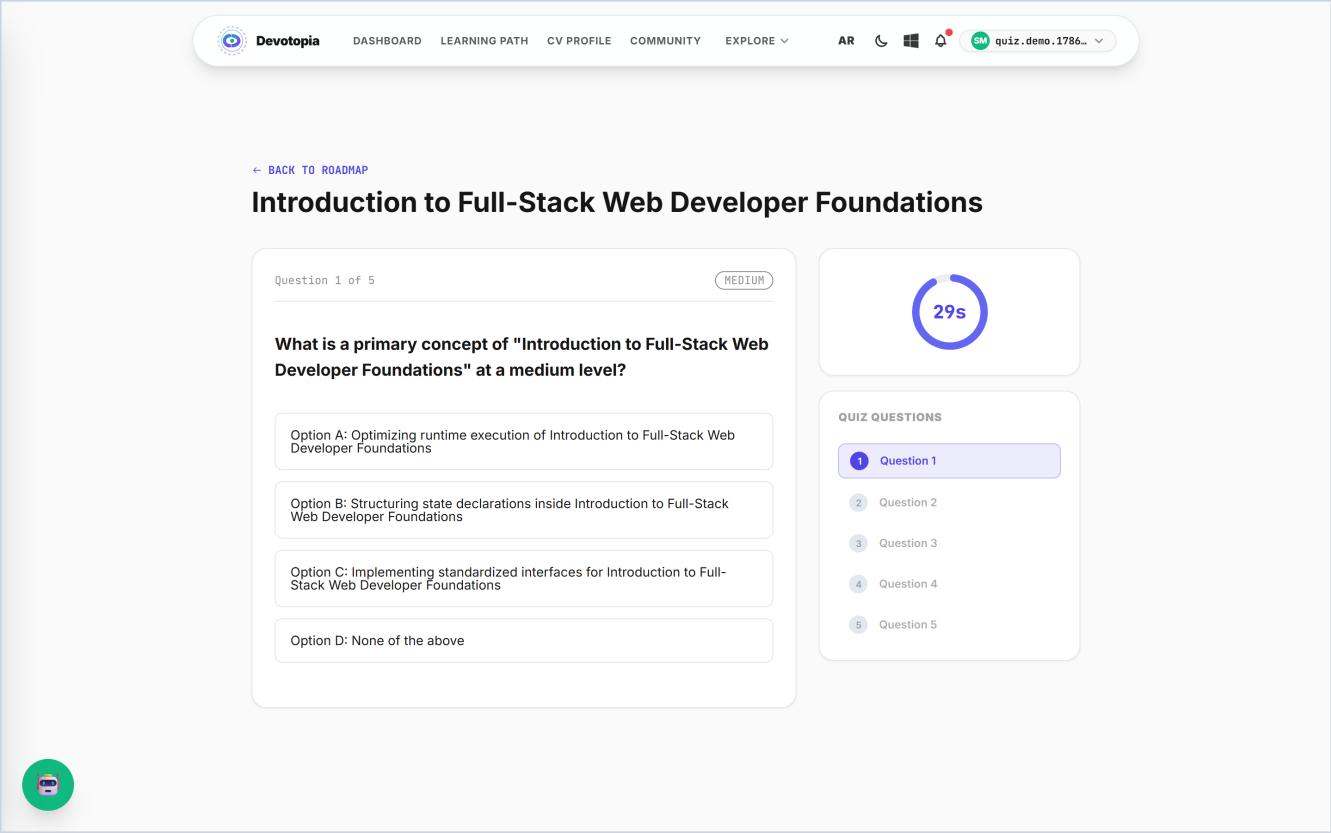
The learning path. Beginner, intermediate and advanced branches with per-node status.



The full roadmap view, including the timeline toggle.

4. The adaptive assessment engine

This is the centre of the project. Five questions per module. The engine watches the last two answers and moves the difficulty of the next question up or down; the final score is then weighted by the difficulty the learner actually faced, so five easy questions and five hard ones do not produce the same mark.



Question 1 of 5. The difficulty badge (MEDIUM) and the per-question timer are visible.

How the difficulty moves

```
// apps/api/src/modules/assessment/assessment.service.ts
// Compute consecutive stats to adjust difficulty for NEXT question
if (answersHistory.length >= 2) {
  const lastTwo = answersHistory.slice(-2);
  const allCorrect   = lastTwo.every((ans) =>  ans.correct);
  const allIncorrect = lastTwo.every((ans) => !ans.correct);
  const currentDifficulty = lastTwo[1]?.difficulty || 'medium';

  if (allCorrect) {
    if      (currentDifficulty === 'easy')   nextDifficulty = 'medium';
    else if (currentDifficulty === 'medium') nextDifficulty = 'hard';
    else                                     nextDifficulty = 'hard';
  } else if (allIncorrect) {
    if      (currentDifficulty === 'hard')   nextDifficulty = 'medium';
    else if (currentDifficulty === 'medium') nextDifficulty = 'easy';
    else                                     nextDifficulty = 'easy';
  } else {
    nextDifficulty = currentDifficulty;
  }
}
```

Two in a row moves the ladder one step. A mixed pair holds position.

How the score is weighted

```
// Calculate weighted score: easy=1, medium=1.5, hard=2
session.answers.forEach((ans) => {
  const weight = ans.difficulty === 'easy' ? 1 : ans.difficulty === 'medium' ? 1.5 : 2;
  totalWeight += weight;
  if (ans.correct) earnedWeight += weight;
});

const finalScorePercent = totalWeight > 0
  ? Math.round((earnedWeight / totalWeight) * 100)
  : 0;
const passed = finalScorePercent >= 70;
```

A hard question is worth twice an easy one, so climbing the ladder is what raises the mark.

What happens on pass, and on fail

The branch after scoring is where the product earns its name. Passing unlocks the dependent modules and checks whether the whole track is now finished. Failing does not simply record a failure — it feeds the missed topics back into curriculum generation.

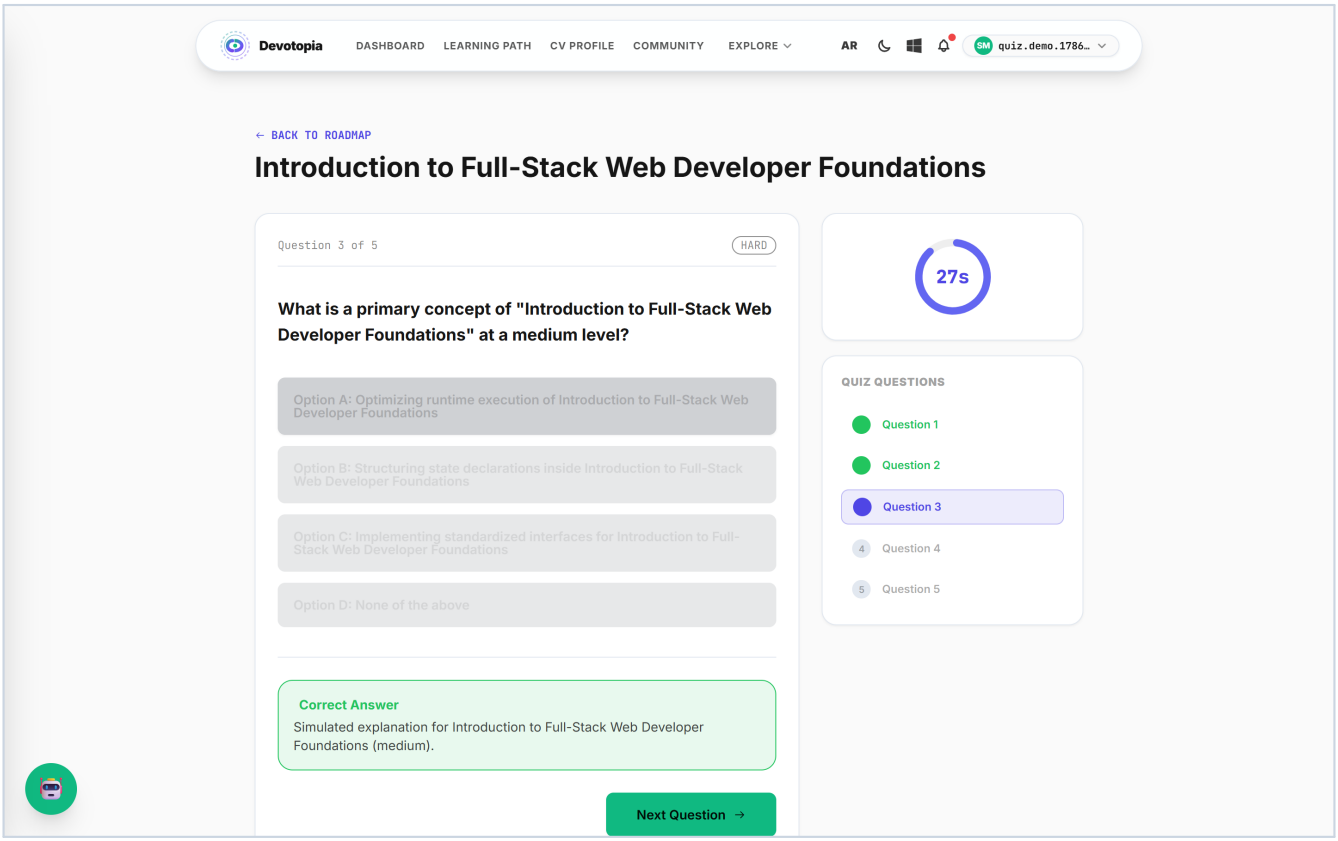
```
// Adaptive outcome: passing unlocks what comes next, failing injects a
// shorter remedial module instead of leaving the learner stuck.
if (passed) {
  await this.unlockNextRoadmapModules(session.userId.toString(), session.moduleId);
  const issuedCert = await this.certService.checkAndIssueCertification(
    session.userId.toString(),
  );
  ...
}

// remedial-node-queue.service.ts
export const REMEDIAL_THRESHOLD = 30; // 30% fail percentage threshold

// assessment.service.ts — recordTopicResult()
result.failPercentage = Math.round((result.failedAttempts / result.attempts) * 100);
if (result.failPercentage >= REMEDIAL_THRESHOLD && !passed) {
  result.status = 'remedial_inserted';
  await this.remedialQueueService.enqueueJob({ userId, topicId, ... });
}
```

The remedial trigger is a threshold on a running fail percentage per topic, not a single bad day.

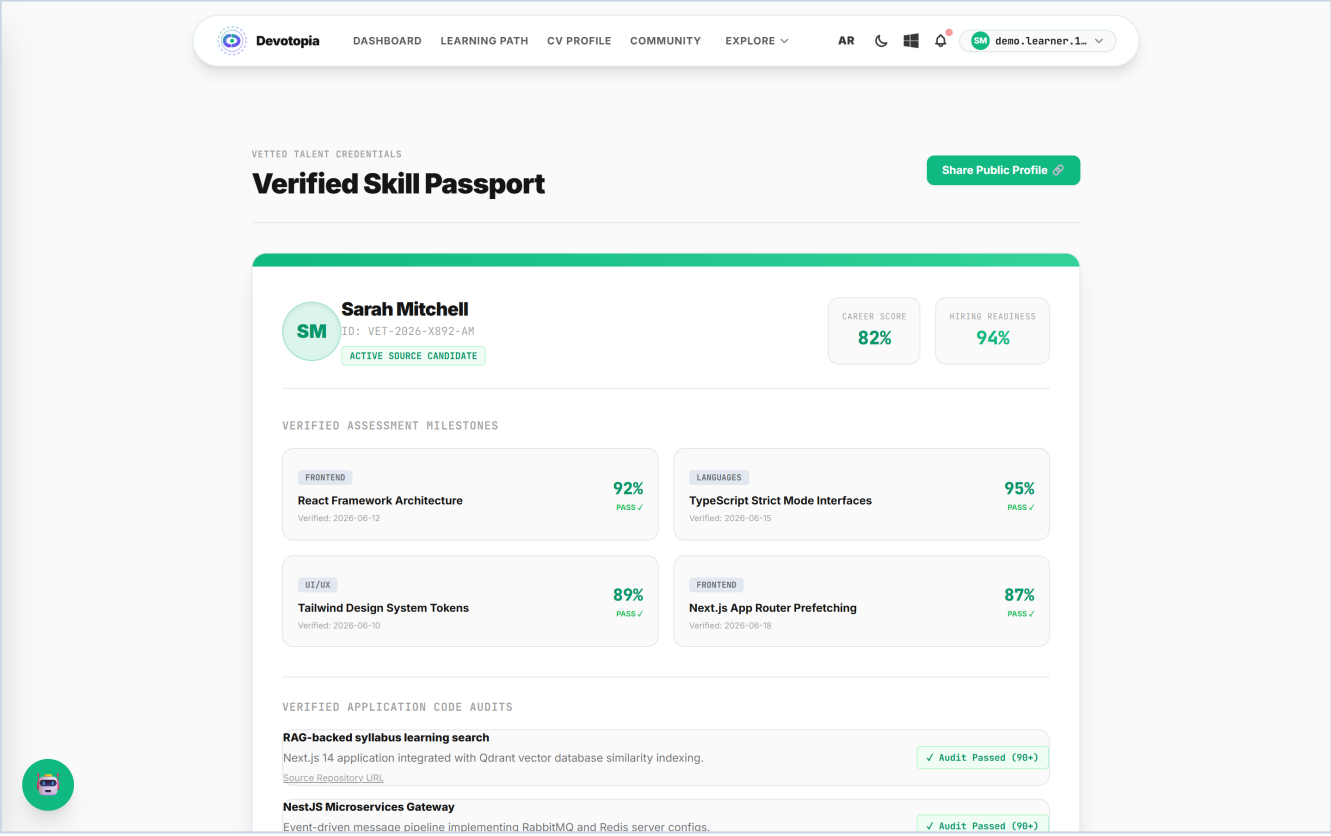
Both rules are covered by unit tests in remedial-trigger.spec.ts, which is possible precisely because the decision is deterministic rather than delegated to a model.



Mid-exam. The sidebar tracks position through the five questions.

5. Turning passes into proof

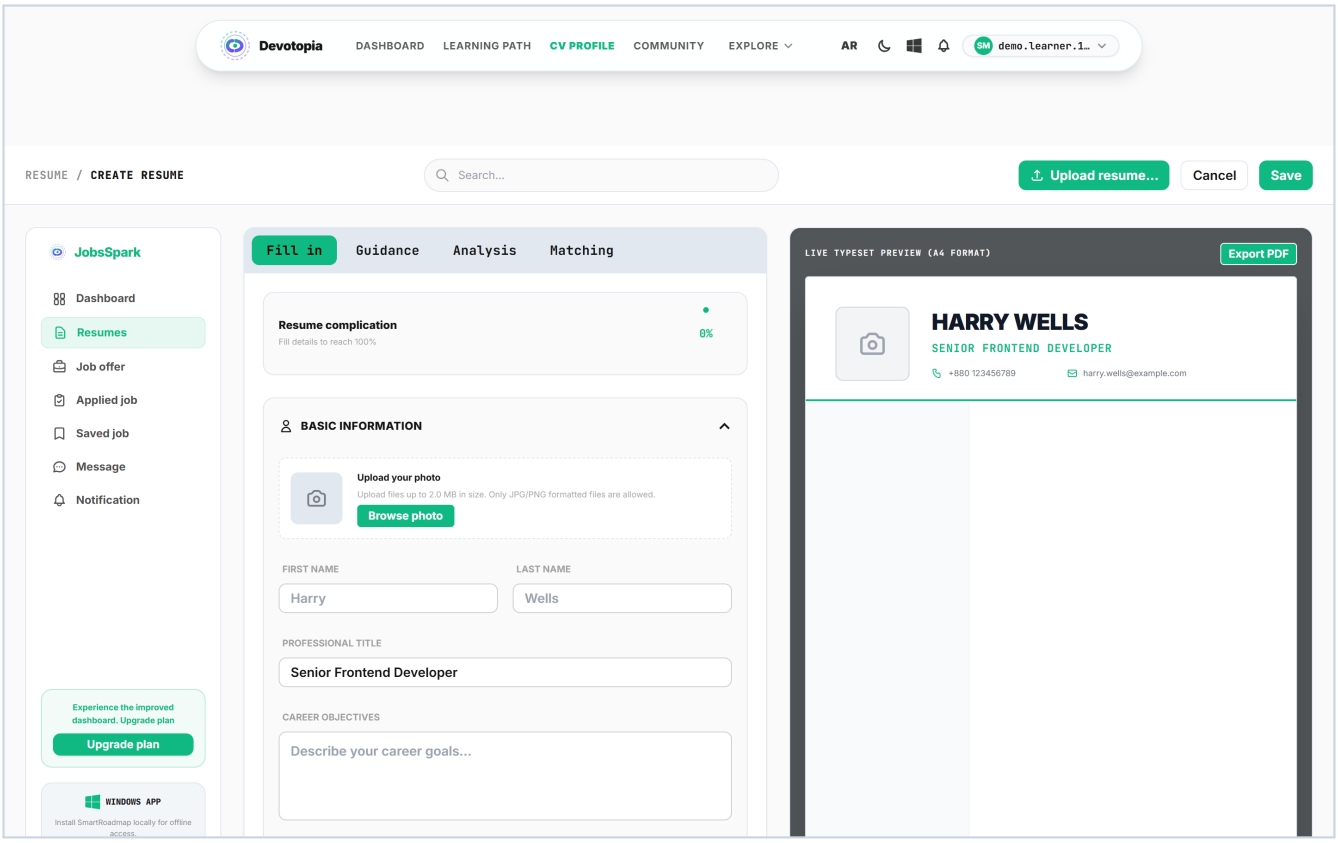
Passed exams accumulate into a passport: a shareable public record of verified milestones with the score and the date each was verified. This is the artefact a recruiter reads, and the same records drive the candidate ranking on the company side.



The verified skill passport, with career score and hiring readiness.

CV BUILDER

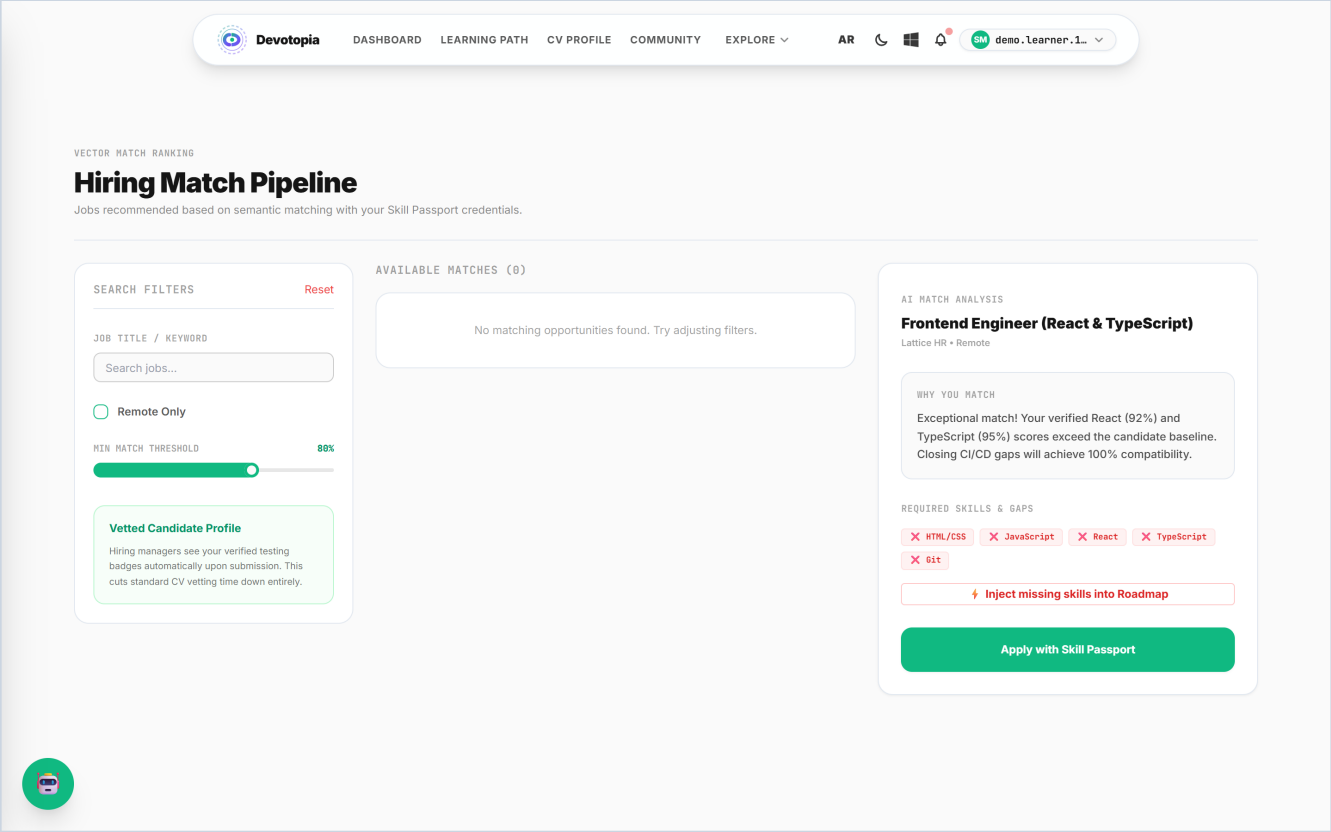
A résumé PDF is parsed into a structured profile — personal details, experience, education, projects, skills — and experience bullets can be rewritten by the model for impact. The parse and the enhancement are separate calls, so a failed enhancement never costs the learner their parsed data.



The CV editor. Structured sections rather than a free-text box.

JOBS AND SKILL GAPS

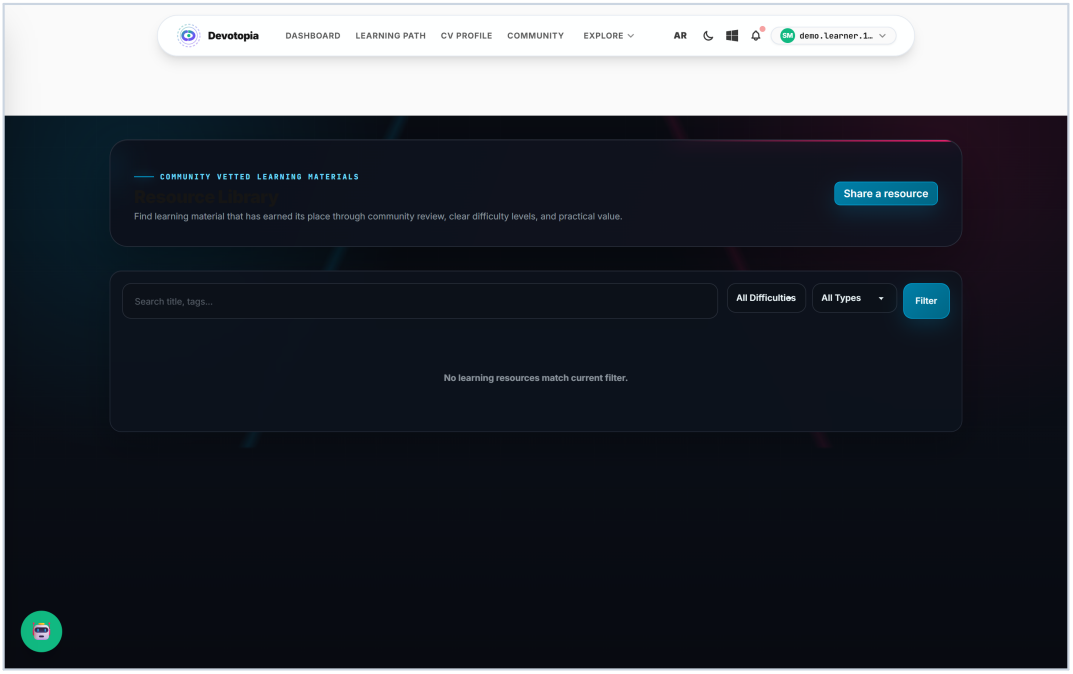
Job matching compares verified skills against a posting's requirements. Where a requirement is missing, it can be injected straight back into the roadmap as new modules — the loop closes back to section 3.



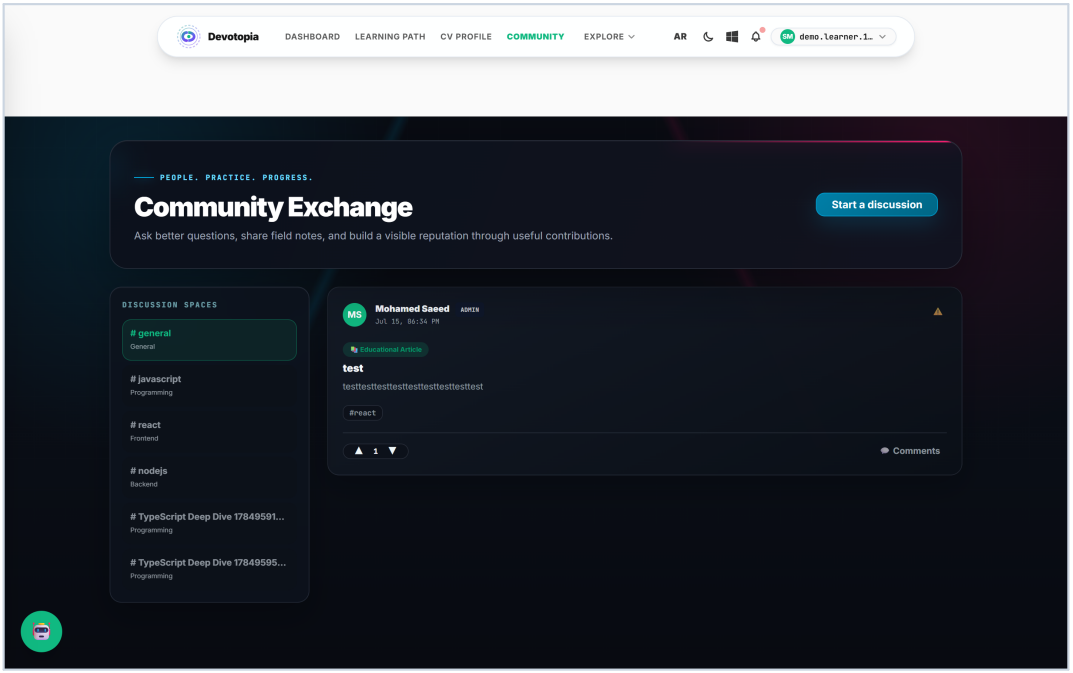
Job matching with the gap comparison.

6. The supporting surfaces

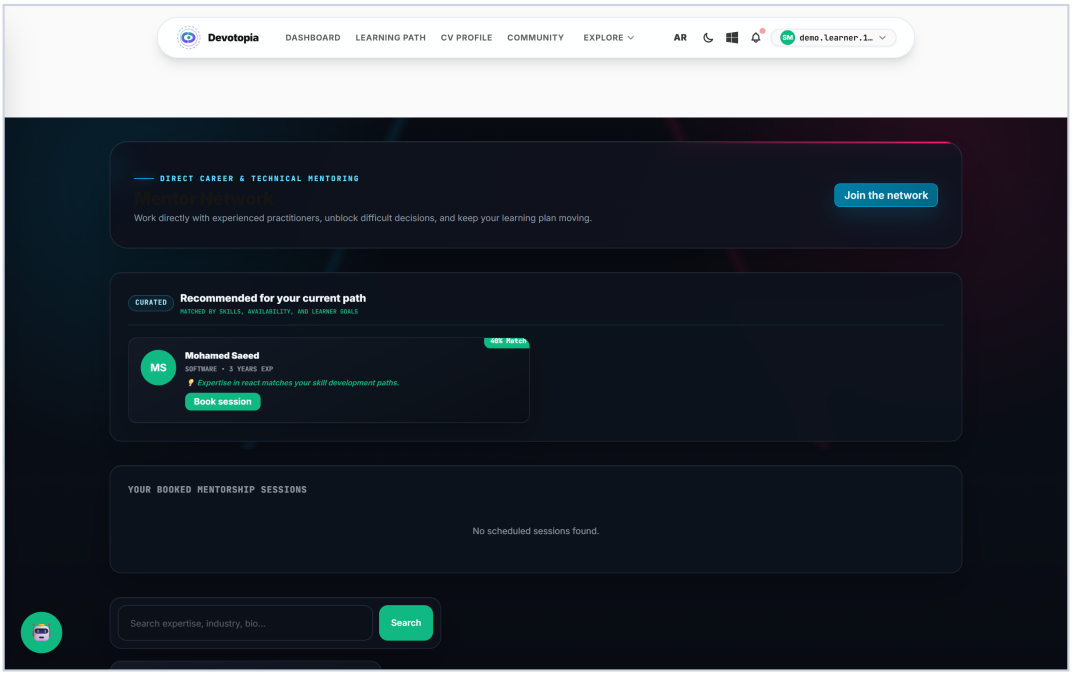
Beyond the core loop the platform carries the surfaces a learner actually needs day to day. Each is a full feature module on the backend, not a placeholder page.



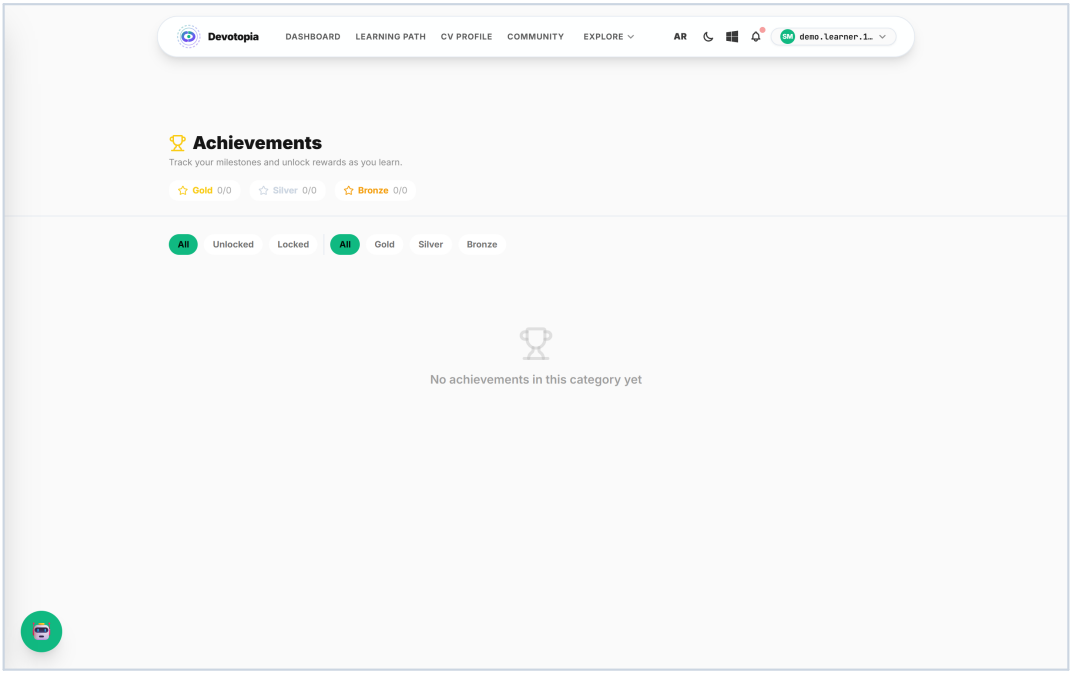
Learning resources, retrieved by semantic search over embedded content.



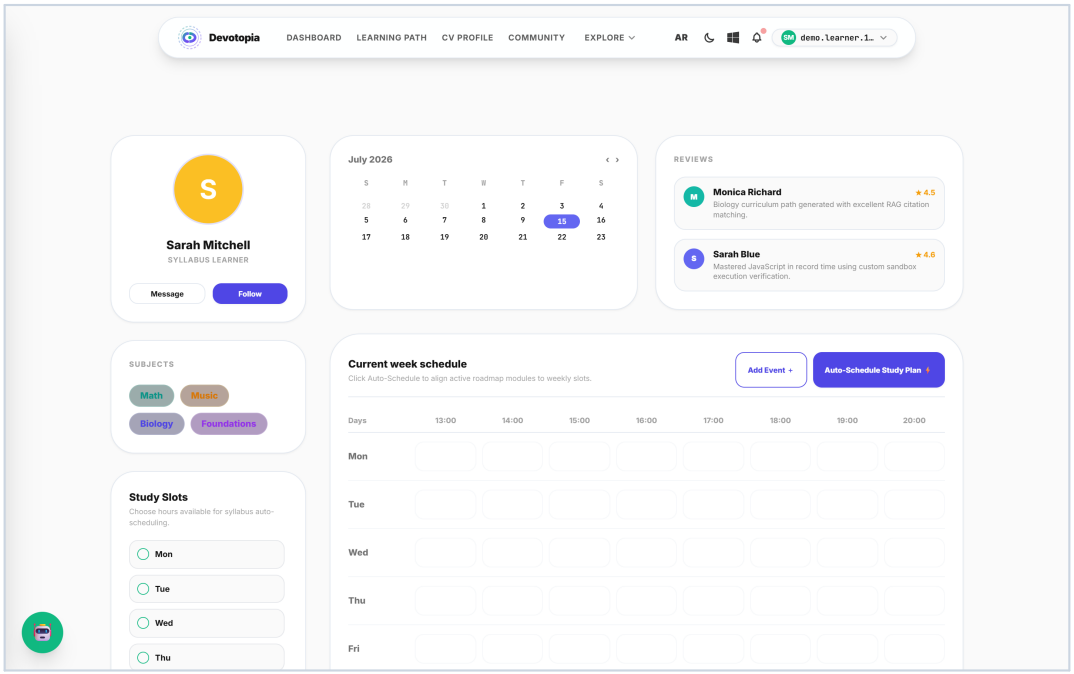
Community discussion spaces.



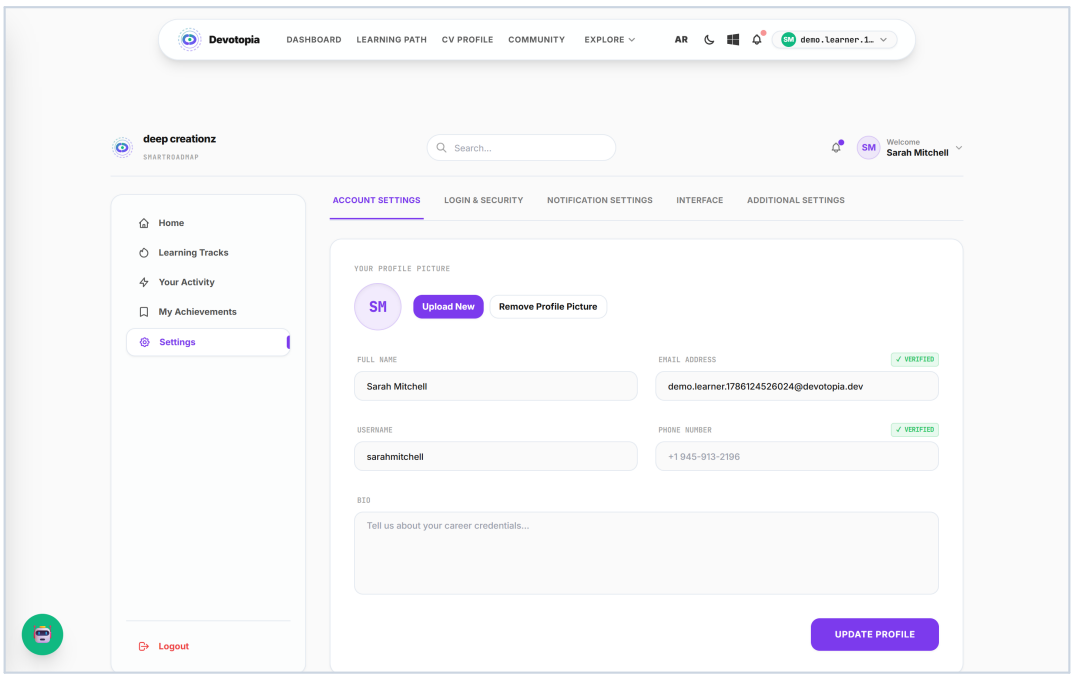
Mentor directory with ratings and session booking.



Achievements — 26 definitions seeded on boot, awarded by event listeners.



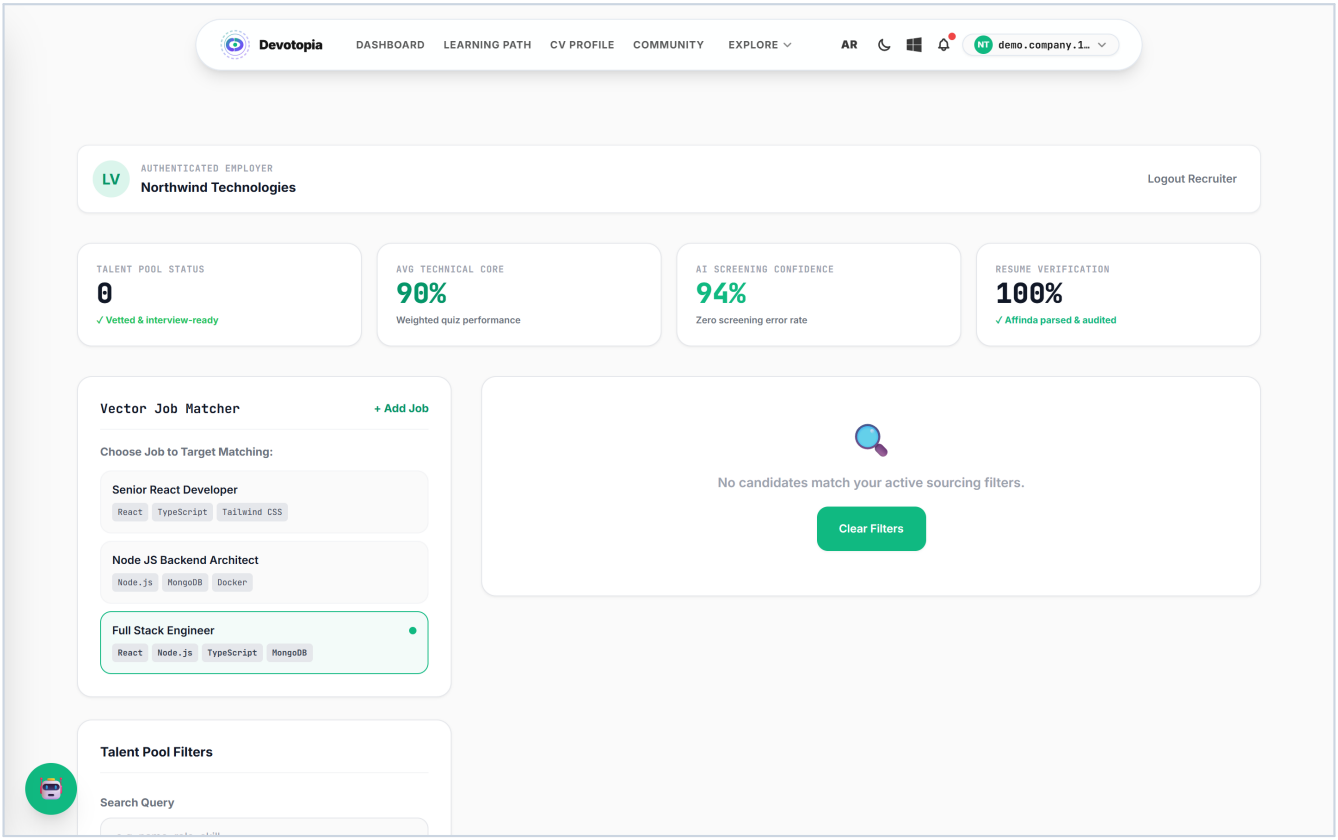
Study calendar and scheduled sessions.



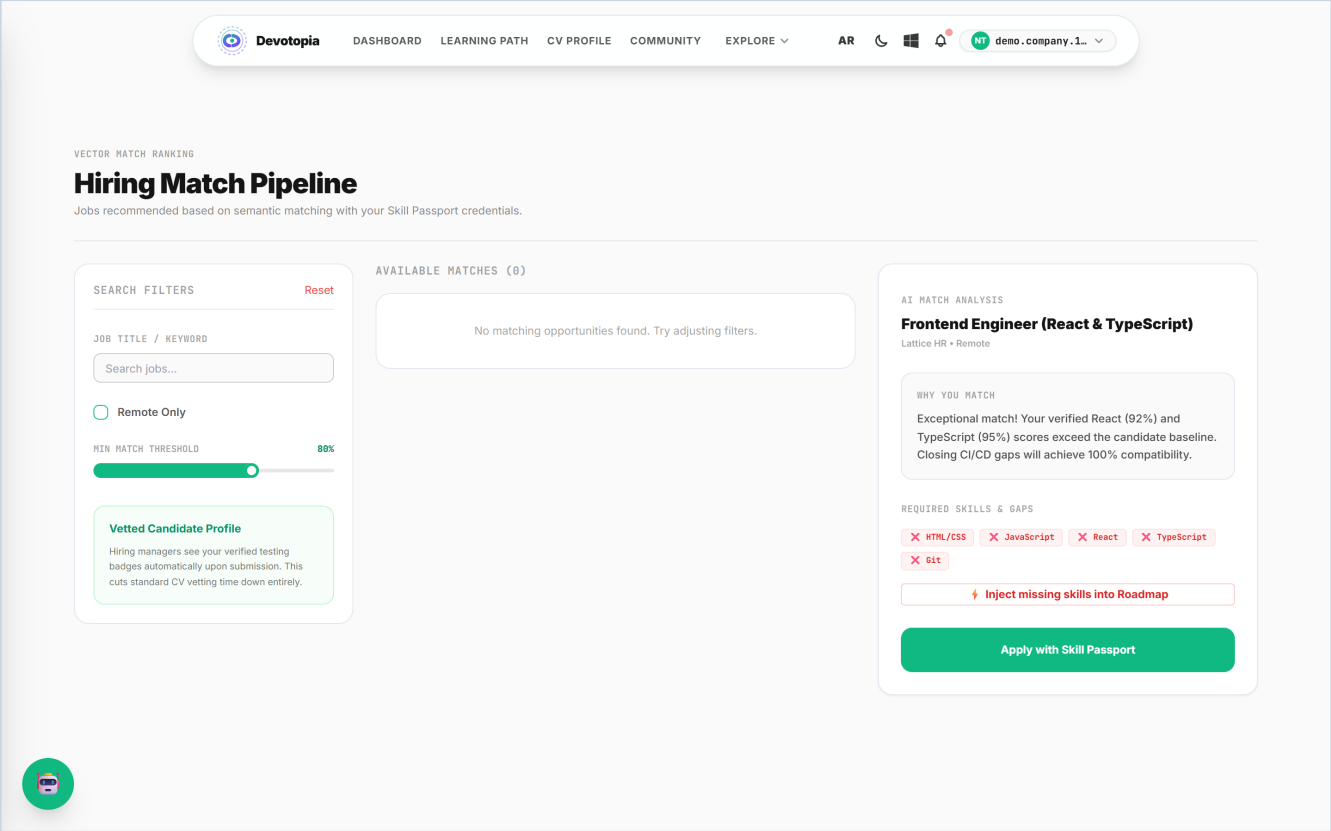
Profile and account settings.

7. The recruiter's side of the platform

A company account sees a different application from the same deployment. Access is not decided in the browser: the role is re-checked on the server for every request, so navigating to a learner route as a company account returns a 403 rather than a rendered page. During this walkthrough that behaviour was observed directly in the API log — a GET on an admin-only route returned 403 for a non-admin session.



The company dashboard.



The candidate pipeline, ranked by verified exam performance.

Where the boundary is enforced

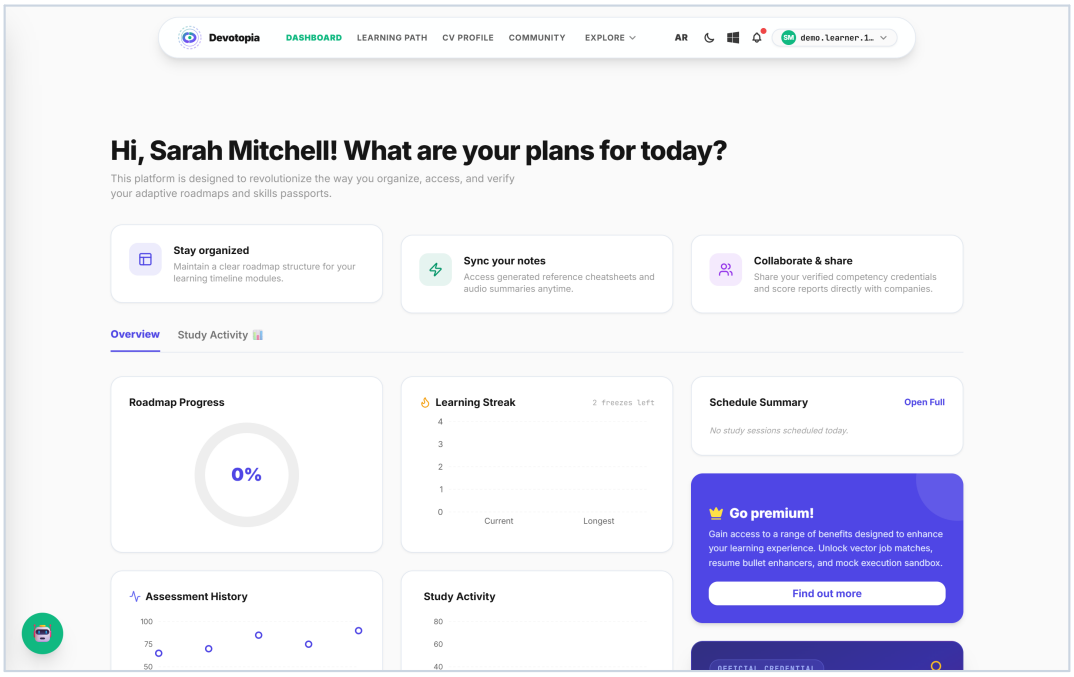
```
// apps/api/src/common/guards/jwt-auth.guard.ts — registered globally
// Deny by default: a route is protected unless it explicitly opts out.
const payload = await this.jwt.verifyAsync(token, {
  secret: this.config.getOrThrow<string>('JWT_SECRET'),
});

// Roles are checked separately, per route, by RolesGuard.
// Ownership of a resource is checked with assertSelfOrAdmin().
```

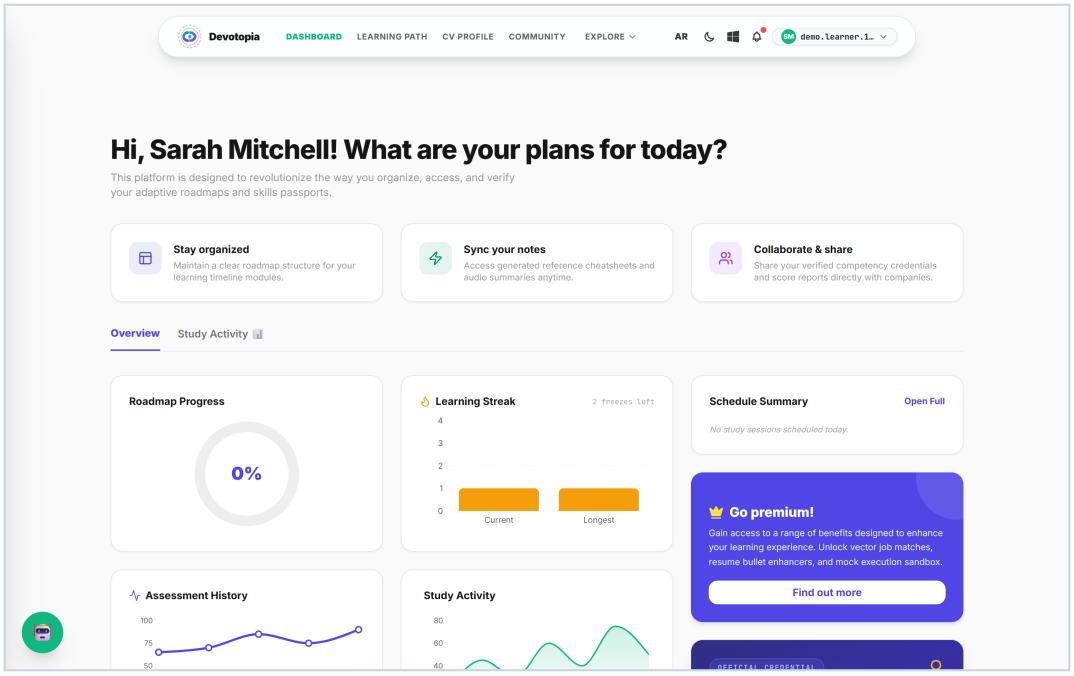
Three independent layers: authenticated, correct role, owns the resource.

8. Two languages, two themes, one layout

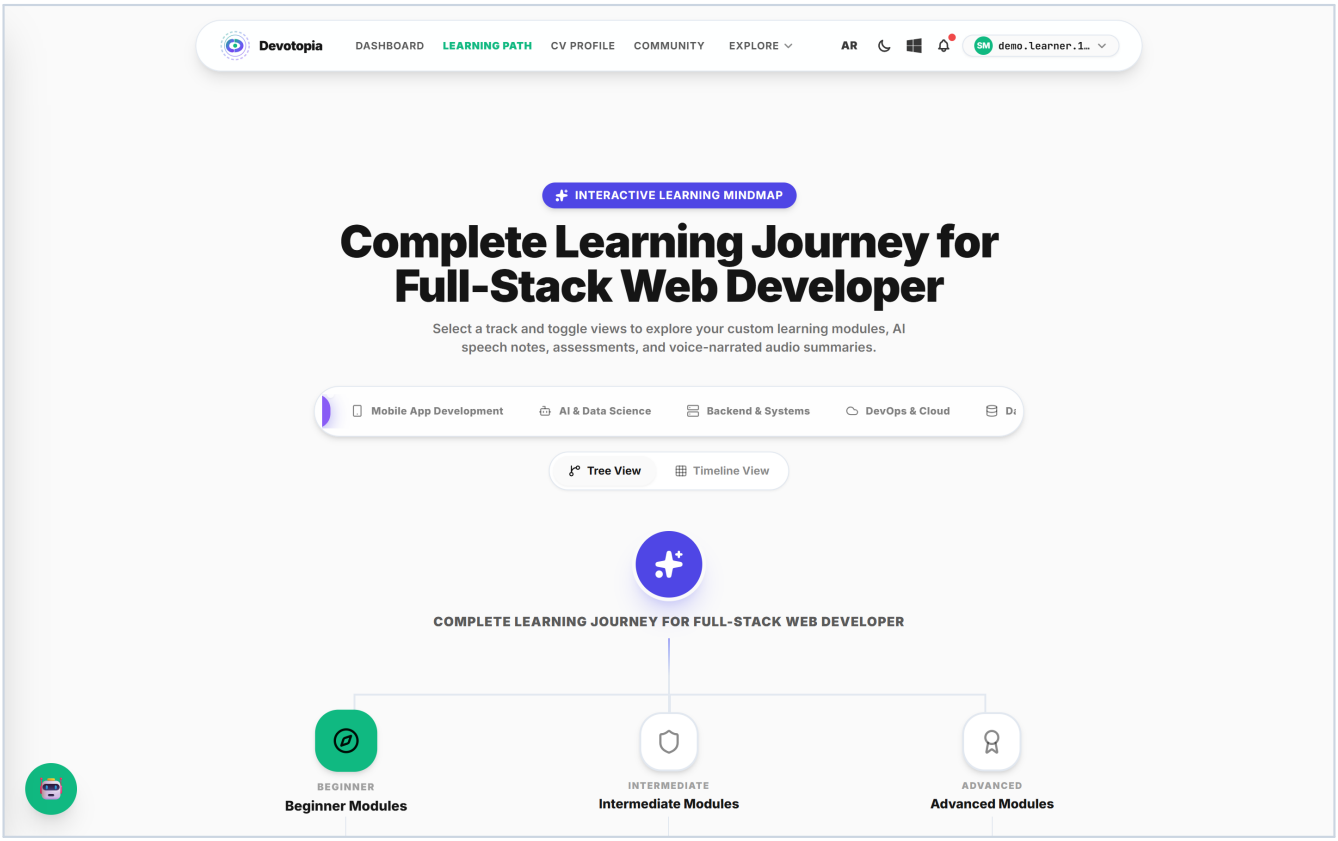
Arabic is not a translation layer bolted on top. Switching locale flips the document direction and the layout follows, because spacing uses logical properties rather than fixed left and right. The same page is shown below in dark mode and in Arabic.



The dashboard in dark mode.



The same dashboard in Arabic, right-to-left.



The roadmap in Arabic — the tree mirrors with the text direction.

9. The AI layer, and what happens when it fails

Every model call goes through one chain. Providers are tried in order and the chain always ends in a deterministic mock, so the product degrades instead of breaking.

```
// apps/api/src/ai/ai-provider.factory.ts
getProvidersChain(): AiProvider[] {
  const chain: AiProvider[] = [];
  const order = ['openai', 'gemini', 'groq', 'huggingface'];
  for (const name of order) {
    const p = this.providers.get(name);
    if (p && p.constructor.name !== 'MockProvider') chain.push(p);
  }
  const mock = this.providers.get('mock');
  if (mock) chain.push(mock);      // last resort, never throws
  return chain;
}

// llm.service.ts — a reply is shape-checked before it is accepted
if (Array.isArray(response.modules) && response.modules.length > 0) {
  return response;
}
// otherwise: log at debug level and fall through to the next provider
```

A malformed reply is treated the same as an outage — try the next provider.

This was not theoretical during capture

While these screenshots were being taken, all three configured live providers rejected every request: OpenAI returned 429 with no credits remaining, and both Gemini and Groq returned invalid-credential errors. The chain fell through to the mock provider and the application kept working — roadmaps generated, exams started, scoring ran. That is the design behaving exactly as intended, and it is why the module titles in the screenshots read as generic placeholders rather than as tailored curriculum text.

The practical consequence is stated plainly: the screenshots demonstrate the full product flow, but the curriculum and question text they contain came from the mock provider, not from a live model. Restoring a valid key on any one of the three providers replaces that text with real generated content, with no code change.

10. Running the project locally

```
# from the repository root
npm install
npm run dev            # turbo runs both apps

# API  → http://localhost:3002
# Web  → http://localhost:3001

npm run smoke          # end-to-end + security suite against a live API
```

The two dev servers used for every screenshot in this document.

What had to be fixed to get it running

A clean checkout did not start. Two independent problems were found and resolved, and both are worth recording because they will affect anyone else cloning the repository.

- Backend dependencies were declared but not installed: @nestjs/event-emitter, bullmq, node-appwrite and web-push. TypeScript reported 11 module-resolution errors and the API never reached bootstrap.
- Frontend dependencies were missing in the same way: framer-motion, recharts, lucide-react, @monaco-editor/react and lineicons. Every authenticated page failed to compile, which is why an earlier capture attempt produced only error screens.
- Two services read configuration with getOrThrow and abort startup when it is absent — AppwriteService needs APPWRITE_ENDPOINT and VoiceAgentService needs ASSEMBLYAI_API_KEY. Placeholder values were supplied at launch rather than written into .env.

After those, both servers start clean: the API compiles with zero TypeScript errors, maps 118 routes, seeds 26 achievement definitions and indexes its static search resources on boot.

Honest status of the environment

Works fully	Auth, roadmap generation, adaptive exams, scoring, unlocking, CV, passport, jobs, community, company pipeline, bilingual UI, themes
Degraded	AI text quality — all three live provider keys are currently rejected, so generated content comes from the mock provider
Not running locally	Redis and Qdrant were not started, so the remedial queue used its in-memory fallback and RAG ran in mock mode

11. What to look at first

If someone reviewing this project has ten minutes, these are the files that carry the ideas rather than the plumbing.

assessment.service.ts	The adaptive ladder, the weighted score, the pass/fail branch and the remedial trigger — the core of the product
ai-provider.factory.ts	The fallback chain, and the decision to always end in a mock
llm.service.ts	How a model reply is shape-checked before it is trusted
jwt-auth.guard.ts	Deny-by-default authentication applied globally
auth.dto.ts	Privilege escalation closed at the validation layer
payment.dto.ts	Server-side price resolution
scripts/smoke-test.mjs	The end-to-end security suite: IDOR, spoofing, rate limits

The through-line in all of them is the same choice: let the model generate content, and keep every decision that determines a learner's outcome in deterministic, testable code.